3.3.3 Number of books and chapters in edited volumes/books published and papers published in national/ international conference proceedings per teacher during last five years (10)

S.No	Academic Year	Papers in National/International Conference Proceedings Per Teacher	Books and Chapters in Edited Volumes / Books Published
1	2020-21	0	04
2	2018-19	31	0
3	2017-18	128	0

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Abstract Proceedings

2nd International Conference on Mathematical Sciences in Engineering Applications (ICMSEA-2017)

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Paper ID: 1032

Exquisite Explanation for the Recognized Conjecture on Fermat Pseudoprimes in Square Form

Narasimha murty Dusi
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Abstract

In this paper we will solve conjecture posed by M.Coman on primes by giving suitable example. We will bring a generalized theorem for square primes, which will deal with the generalization of Fermat Pseudoprime, which gives infinitely many pseudoprimes for different base system.

Keywords: Fermat little theorem; Fermat pseudoprime; Weifrich primes;

* * *

Paper ID: 2001

Effect on Compressive Strength of Concrete by Substitution Of Fine Aggregate With Blast Furnace Slag

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Dr.N.Victor Babu, Professor, Department of Civil Engineering, BABA Institute of Technology & Sciences, Visakhapatnam.

Abstract

The iron industries produce a huge quantity of blast furnace slag as by-product, which is a non-biodegradable waste material from that only a small percentage of it is used by cement industries to manufacture cement. In present inspection Blast Furnace Slag from Vizag-Steel Plant has been utilized for finding its appropriateness as a Fine Aggregate in concrete. Replacement of some natural aggregates with Slag would lead to abundant environmental benefits. It is interesting that the Blast Furnace Slag (BFS) can be used as a fine aggregate in concrete. Based on overall experimental results, it could be advisable that the Slag could be effectively utilized as fine aggregate in concrete practice.

Keywords: Blast Furnace Slag (BFS), Fine Aggregate; Concrete; Environmental benefits; Compressive Strength; Slag Aggregate; Iron Industries;

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Paper ID:2004

Study of normalized difference built-up (NDBI) index in automatically mapping urban areas from Landsat TM imagery.

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Abstract

The Visakhapatnam city lies on the eastern part of India and the urban sprawl in Visakhapatnam town was analyzed using multi-temporal Landsat TM data from 2005 to 2015. Spectral indices namely Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI) and Built-up Index (BUI) were generated from the Landsat TM bands covering visible Red (R), Near Infrared (NIR) and Short Wave Infrared (SWIR) wave-length regions. Spectral variations in built-up, open spaces, urban vegetation and water areas were studied by generating two-dimensional spectral plots of NDBI and BUI. Remote sensing images are mainly used for monitoring and detecting the land cover changes that occur frequently in urban and sub-urban areas as a consequence of incessant urbanization. The conversion of satellite imagery into land cover map using the existing methods of manual interpretation and parametric image classification digitally is a lengthy process. Normalized Difference Built-up Index (NDBI) which is used for mapping built-up areas automatically. This method main advantage is the unique spectral response of built-up area and other land covers. Built-up area can be effectively mapped through arithmetic manipulation of re-coded Normalized Difference Vegetation Index (NDVI) and NDBI images derived from TM imagery. This NDBI method is applied to Vishakaptanam Area in this paper and the mapped area results at accuracy of 93.9% indicates that it can be used to fulfill the mapping objective reliably and efficiently. Comparing NDBI with the maximum likelihood classification method, this NDBI method is able to serve as a worthwhile alternative for quickly and objectively mapping built-up areas.

Key words: Difference Vegetation Index, Normalized Difference Built-up Index, Built-up Index and Spectral variations.

Paper ID: 2006

Soil Stabilization Using Plastics

Ch.Grace Evangelin M.Mohan

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Abstract

It reflects that plastic wastes can be used in stabilization of soil which is concluded from various tests conducted on fiber reinforced soil with varying fiber content. In this study, different means of plastic waste as shopping bags which are locally available are used so as a reinforcement to perform unconfined compressive test while mixing with soil for improving engineering performance of sub grade soil. In this the plastic strips which are collected for stabilization of soil were mixed randomly with the soil. With this a series of tests were carried out on randomly reinforced soil with varying percentage of plastic strips starting from 0, 0.5%, 1.5% and 2.5%. Use of plastic in soil in an appropriate amount really aids in improving the strength of soil and can be useful as sub grade soil which can be used in pavement construction.

Keywords: Stabilization; Unconfined Compressive Strength; Fiber Reinforced Soil; Soil Properties; Shear Strength.

Paper ID: 2008

WATERSHED PRESERVATION AND CONTROL USING GIS

V.Naga Sai Sindhuja, CH.Ramesh Naidu, Murali Krishna Gurram

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Abstract

Remote sensing (RS) and Geographical Information System (GIS) data provides timely, accurate and reliable information at definite intervals in a cost effective manner and these tools are helpful in mapping and monitoring of natural resources available for proper utilization.The Meghadrigedda watershed is selected for the present study and has a gross capacity of 1169 Mc. Ft. For the identification of morphological features and analyzing their properties of the Meghadrigedda basin various GIS and image processing techniques have been adopted. The linear and aerial aspects of the basin were calculated and computed. It is 7th order drainage basin and drainage pattern is dendritic type. The results obtained from watershed delineation and prioritization has wider application in preservation of the watershed.

Keywords: GIS; Watershed; Delineation; Drainage Parameters; Preservation.

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Paper ID: 2008

WATERSHED PRESERVATION AND CONTROL USING GIS

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WATERSHED PRESERVATION AND CONTROL USING GIS

V.Naga Sai Sindhuja*
Dr. N. Victor Babu**
CH.Ramesh Naidu***
Murali Krishna Gurram****

Abstract

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Keywords:

GIS;
Watershed;
Delineation;
Drainage Parameters;
Preservation.

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1.Introduction

Basin morphometry is the mathematical quantification of various drainage basin characteristics. Land is one of the important natural resource and it should be conserved to mitigate the increasing demand of land and water resources (Panhalkar S.S et al, 2012). In India 70% of the people depend on agriculture directly or indirectly. The analysis of the drainage basin is important in assessing the ground water potential and ground water management. The study on morphometric analysis using remote sensing and GIS techniques was carried out by (Sangita Mishra et al, (2010); Biswas et al. 1999; Panhalkar S.S et al, 2012; Subodh Chandra Pal and Gopal Chandra Debnath, 2012; Jagadeeswara Rao, Harikrishna P and Suryaprakasa Rao.B, 2006).

The Meghadrigedda watershed is the area enclosed between Northern latitude and Eastern longitude, covering an area of 453.40 sq. km falling in Survey of India (SOI) toposheet Nos: 65 O/1, O/2&O/3, O/5 and O/6 on 1:50,000 scale. Meghadrigedda reservoir was formed across Meghadrigedda and Naravagedda just 400 mts above their confluence to supply drinking water to the people of Visakhapatnam city at the rate of 8Million gallons per day. The 8MGD of water is being drawn from the reservoir and distributed to various industries and industrial colonies by Visakhapatnam Municipal Corporation.

Meghadrigedda reservoir has a gross capacity of 1169 Mc.Ft and the flood bank was formed on the left side of the reservoir to protect the Howrah-Visakhapatnam railway line and the Chintalagraharam village. The average annual rainfall is around 110cm and occurs in the months of June to November. December and January are coldest months in this region and April and May are hottest months. The area enjoys sub tropical climate and the temperature ranges from 14 to 22° C in December and 33-45°C in May. Relative humidity is high in day time and varies between 70-80% throughout the year.

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****Head, GIS Technology & Applications, Xinthe Technologies Pvt. Ltd, Visakhapatnam.

Paper ID: 2011

Phytoremediation(Plants as reactors)

M.Mohan Ch. Grace Evangelin V. N.S. Sindhuja M. Chiranjeevi A. Uday Kumar
M. Mohan, Assistant professor, BITS VIZAG. Email: mmohan262@gmail.com

Abstract

Now -a- Days we have two types of treatment processes they are physiochemical process and biologic process.in biological process ASP, trickling filters, RBC, oxidation ponds, and phytoremediation. present phytoremediation is in the early stage of commercialization for treatment of wastewater containi organic and inorganic pollutants, and in the future it may provide a low cost option. In phytoremeditic we use plants for treating waste water. In that roots are influent collectors and leaves are product delive In this we have six reactions takes place they are phytoextraction, rhizofiltration, phytostabilizati phytodegradation, hizodegradation and phytovolatilization.Those plants also help prevent wind, rai and groundwater from carrying pollutants away from sites to other areas. Wetland construction will he to remove or reconciliation of wastewater to environment from hazardous pollutants. Yellow or Whi Water Lillis, Alfalfagrass, these type of plants are useful to remove organic pollutants.Brassica fami (Indian Mustard & Broccoli),Tomato plant, sunflower these are useful to remove inorganic pollutant

* * *

Paper ID: 2012

Experimental investigation on partial replacement of Cement with GGBFS and Sand with Bottom Ash in concrete

B.V.Ramanamurthy, Dr.N.Victor Babu Neeli, Harikrishna, B.V.Ramanamurthy

Assst.Professor, Department of Civil Engineering, Baba Institute of Technology and Sciences, Visakhapatnam. Email: vrmurthyb@gmail.com

Abstract:

Nowadays Cement is major constituent material in construction industry and Lime and Silica produces the Cement.Without these natural materials production of cement is very tedious task.Moreover the emission of Carbondioxide(CO_2) is more in the production of Cement.It causes environmental pollution globally.Sand is also naturally available material ,it gives good strength to the concrete.But environmental degradation happened by scarcity of natural sand.The main motto of this work is to find alternate materials for partial replacement of Cement and sand in concrete to obtain required strength as well as to minimize degradation of natural materials.In this work Cement is partially replaced by Ground GranulatedBlast Furnace slag(GGBFS) and sand is Bottom Ash(BA) to achieve required compressive strength.By this investigation,replacement of 10% Cement and 10% sand in concrete with and without adding of admixture gave sufficient characterstic compressive strength of concrete for 28 days curing period.

Key words: Characterstic compressive strength GroundGranulatedBlast Furnace slag(GGBFS) Bottom Ash(BA).

* * *

* * *

Paper ID: 2012

Experimental investigation on partial replacement of Cement with GGBFS and Sand with Bottom Ash in concrete

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Paper ID: 2013

Comparative Study On End Moments Regarding Application Of Analytical Methods And Staad.Pro Software

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Email: chirum4u@gmail.com

Abstract

Analysis of portal frames involves lot of complications and tedious calculations by conventional methods. To carry out such analysis it is a time consuming task. The Moment Distribution Method & Kani's Method for analysis of portal frames can be handy in approximate and quick analysis so as to get the detailed estimates ready. In this work, these two methods have been applied only for uniform distributed loading conditions. This paper mainly deals with the comparative analysis of the results obtained from

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COMPARATIVE STUDY ON END MOMENTS REGARDING APPLICATION OF ANALYTICAL METHODS AND STAAD.Pro SOFTWARE

Chiranjeevi.M*
Udaykumar.A**
S.V.Saikumar***

Abstract

Analysis of portal frames involves lot of complications and tedious calculations by conventional methods. To carry out such analysis it is a time consuming task. The Moment Distribution Method & Kani's Method for analysis of portal frames can be handy in approximate and quick analysis so as to get the detailed estimates ready. In this work, these two methods have been applied only for uniform distributed loading conditions. This paper mainly deals with the comparative analysis of the results obtained from the analysis of single bay portal frame when analyzed manually by using STAAD.Pro Software separately. The result obtained from manually is mostly matched with the results obtained from STAAD.Pro software.

Keywords:

Portal Frame;
Moment Distribution Method;
Kani's Method;
STAAD.Pro Software.

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Introduction

For the analysis of portal frames, Moment distribution method and Kani's method of analysis are mainly used, which allows the engineer to analyse frames easily and design it economically. The research is concluded by evaluating a selection of portal frame, with practical dimensions in order to substantiate the conclusions as stated below.

This paper presents the analysis of portal frame, considering mainly the case of single bay, which is the most common in practice. Portal frames are very efficient and economical when used for single-storey buildings, provided that the design details are cost effective and the design parameters and assumptions are well chosen. Portal frames consists of a horizontal beam resting on two columns are single storey with pitched or flat roof (Fig.1). The junction of the beam with the column consists of rigid joints and fixed at the base. The main objective of this paper is analysing the portal frame by manually and compare their results with the help of STADD.Pro software.

2. History and Methods of Analysis

2.1Structure

In engineering and architecture, a structure is the combination of two or more basic structural components connected together in such a way that they serve the user functionally and carry the loads arising out of self and super-imposed loads safely without causing any problem of utility.

2.2 Structural Analysis

Structural analysis deals with study and determination of forces in various components of a structure subjected to loads. As the structural system as a whole and the loads acting on it may be of composite nature

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Structural analysis deals with study and determination of forces in various components of a structure subjected to loads. As the structural system as a whole and the loads acting on it may be of composite nature

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COMPARATIVE STUDY ON END MOMENTS REGARDING APPLICATION OF ANALYTICAL METHODS AND STAAD.Pro SOFTWARE

Chiranjeevi.M*
Udaykumar.A**
S.V.Saikumar***

Abstract

Analysis of portal frames involves lot of complications and tedious calculations by conventional methods. To carry out such analysis it is a time consuming task. The Moment Distribution Method & Kani's Method for analysis of portal frames can be handy in approximate and quick analysis so as to get the detailed estimates ready. In this work, these two methods have been applied only for uniform distributed loading conditions. This paper mainly deals with the comparative analysis of the results obtained from the analysis of single bay portal frame when analyzed manually by using STAAD.Pro Software separately. The result obtained from manually is mostly matched with the results obtained from STAAD.Pro software.

Keywords:

Portal Frame;
Moment Distribution Method;
Kani's Method;
STAAD.Pro Software.

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Introduction

For the analysis of portal frames, Moment distribution method and Kani's method of analysis are mainly used, which allows the engineer to analyse frames easily and design it economically. The research is concluded by evaluating a selection of portal frame, with practical dimensions in order to substantiate the conclusions as stated below.

This paper presents the analysis of portal frame, considering mainly the case of single bay, which is the most common in practice. Portal frames are very efficient and economical when used for single-storey buildings, provided that the design details are cost effective and the design parameters and assumptions are well chosen. Portal frames consists of a horizontal beam resting on two columns are single storey with pitched or flat roof (Fig.1). The junction of the beam with the column consists of rigid joints and fixed at the base. The main objective of this paper is analysing the portal frame by manually and compare their results with the help of STADD.Pro software.

2. History and Methods of Analysis

2.1Structure

In engineering and architecture, a structure is the combination of two or more basic structural components connected together in such a way that they serve the user functionally and carry the loads arising out of self and super-imposed loads safely without causing any problem of utility.

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Paper ID: 2014

Technical Advancement By Rice Husk Ash In Low Cost Housing Construction

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Abstract

The improvement of concrete technology can able to decrease the pollutants on environment and u of natural resources. This paper explains the usefulness of using Rice husk ash (RHA) in con production as a partial replacement of cement. After replacement, this can improve the strength as we durability properties of concrete. RHA has high reactivity and pozzolonic property and this pro depends on temperature. Concrete mixtures were produced and tested both with RHA and without l replacement in terms of compressive strength and results have been compared between both of t 30% replacement of cement with RHA has been done in all cases. Soil stabilization and ground impro techniques have become a important issue in construction field now a days. The common soil stabiliz methods are becoming more costly because of high cost of stabilizing agents like lime, cement etc cost of the stabilization may be decreased by proper replacement of RHA with suitable proportion replacement with RHA and also different tests have been conducted to check the effectiveness of RH the soil stabilization aspect. So that RHA usage can reduce the cost of the total construction.

Keywords: Rick husk ash (RHA); Soil stabilization; Replacement of Cement; Concrete techno Pozzolonic

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Paper ID: 2014

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* * *

Paper ID: 2015

Use of Recycled Materials In Concrete Construction

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Abstract

Use of recycled aggregate in concrete can be useful for environmental protection. Recycled aggreg are the materials for the future. The application of recycled aggregate has been started in a large num of construction projects of many European, American, Russian and Asian countries.This paper rep the basic properties of recycled fine aggregate & also compares these properties with natural aggrega Similarly the properties of recycled aggregate concrete are also determined. Basic concrete proper

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USE OF RECYCLED MATERIALS IN CONCRETE CONSTRUCTION

Mr.Nanubilli Ramu*
Dr.N.Victor Babu**

Abstract

Use of recycled aggregate in concrete can be useful for environmental protection. Recycledaggregates are the materials for the future. The application of recycled aggregate has been started in a large number of construction projects of many European, American, Russian and Asian countries.This paper reports the basic properties of recycled fine aggregate & also compares these properties with natural aggregates. Similarly the properties of recycled aggregate concrete are also determined. Basic concrete properties like compressive strength,split tensile strength, flexural strength and workability etc. are explained here for different combinations of recycled aggregate with natural aggregate. The concrete mixes were designed using IS 10262-2009 .fine aggregate is partially replaced with 10%,20%,30% and cement is partially replaced with silicafume of 0%,10%and 15% respectively.

Keywords:

Rubber agregate;
Normal aggregate;
Cement;
Silica fume.

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1. Introduction

Waste materials are common problems in modern living. Waste accumulates from a number of sources including domestic, industrial, commercial and construction. These waste materials have to be eventually disposed of in ways that do not endanger human health. In light of this, waste minimization is increasingly seen as an ecologicallysustainable strategy for alleviating the need for the disposal of wastematerials,which is often costly, time and space consuming, and can also have significant detrimental impacts on the natural environment.The use of recycled materials is often cheaper for the consumers of the end product. Hence, there is also an economic justification for promoting its use.

The use of recycled materials generated from transportation, industrial, municipal and mining processes in

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Paper ID: 2021

Strength Improvement of M50 grade concrete as a partial replacement of fine aggregate with Stone Crusher Dust.

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Abstract

Concrete is used widely in all types of constructions ranging from small buildings to large structures like dams or reservoirs. It is the most widely used construction material to improve the strength& quality of structural elements. Sand is one of the important ingredients in the making of concrete but the availability of sand is a tedious task. For the protection of environment, almost all local governments have to be taken a decision to minimize sand excavation for utilization in construction. The current study an attempt has been made to studying the behavior of the concrete made by partial replacement of sand with Stone Crusher Dust (SCD) using admixture. The design mix prepared for M53 grade concrete as per IS Code and fine aggregate has been replaced by Stone Crusher Dust at 10%, 20%, 30%, 40% and 50% by weight. The strength tests have been made to study for 3, 7 and 28days curing period.

Key words: Compressive strength, Sand Crusher Dust (SCD), Admixture.

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Paper ID: 2021

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PAPER ID: 2025

Comparative Study On End Moments Regarding Application Of Slope Deflection, Flexibility Matrix Method And Staad.Pro Software

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Abstract

In the analysis of continuous beam, there are many complexities and monotonous calculations involved in traditional methods. End moment is one of the main considerations from the structure design. In this present study determine the end moment's two various procedures i.e. Slope deflection and flexibility matrix method have been used for continuous beam. The present work shows that the End moments determined from these methods have almost same result when compared with STAAD.Pro Software. The results determined from slope deflection method are quite accurate value then the compare with the STAAD.Pro

Keywords: Continuous Beam; Slope Deflection Method; Flexibility Matrix Method; STAAD.Pro.

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Comparative Study On End Moments Regarding Application Of Slope Deflection, Flexibility Matrix Method And Staad.Pro Software

A.Udaykumar, M.Chiranjeevi, S.V.Saikumar

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Keywords: Continuous Beam; Slope Deflection Method; Flexibility Matrix Method; STAAD.Pro.

* * *

Paper ID 4014

Determination of Thermophysical Properties of Nano Hybrid Refrigerants to be used in Commercial Refrigeration System

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Abstract

In present scenario, the need of increased cooling rate in commercial applications of a Refrigeration system is high. However, the complications in power consumption and lesser COP are still challenges to the researchers. Further, environmental impacts of refrigerant is also need to be considered. Hence, suitable refrigerant with increased thermophysical properties is need to be identified for increasing the performance of commercial refrigeration system. From the literature, it is reported that the use of mixed refrigerants in refrigeration system can enhance cooling capacity. However, mixed refrigerants with nano particles are yet to be studied in commercial refrigeration applications. In this context, the thermophysical properties of nano hybrid refrigerants are investigated by using R-22 and R-12 with a mass fraction of 50% each by volume concentration of nano particle TiO2 from 1% to 5%. Different mathematical correlations which are developed in the past are used to determine the thermophysical properties of hybrid nano refrigerants. The thermo-physical properties of hybrid refrigerants such as Specific heat, Density, Thermal conductivity and Viscosity are determined by varying temperature from 300-350K with an temperature increment of 5K. From the results, it is observed that by nano particles dispersions in mixed refrigerants increased the thermo-physical properties. Hence, it is concluded that nano hybrid refr can enhance the cooling capacity of commercial refrigeration

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Paper ID 4014

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Paper ID 4030

Experimental Investigation and Optimization of wear rate in Al6062-SiC-Flyash Metal Matrix Composite

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Abstract

Aluminium Metal Matrix composites is a relatively attractive material for automobile, aerospace and other engineering application due to its mechanical and tribological properties. To improve wear resistance and mechanical properties has led to design and selection of newer variants of the composite. The present investigation deal with the study of wear behavior of Al6062-SiC-flyash MMCs for varying reinforcement content, applied load, sliding speed, and distance. Aluminium MMCs reinforced with three different percentage of reinforcement 3, 6, 9% wt. SiC and 0.5, 0.75, 1.00% wt. flyash prepared by stir casting method. Wear test was performed by using "pin on disk" apparatus. A plan of experiment based on L27 Taguchi orthogonal array is used to acquire the wear data. An analysis of variance is employed to investigate the influence of four controlling p sliding distance on dry sliding wear of the composites. It is observed that SiC content, sliding speed and normal load significantly affect the dry sliding wear. The optimal combina minimum wear. The microstructure study of worn surface indicates nature of wear to be mostly abrasive.

Keywords: Metal matrix composites; Al-SiC-flyash alloy; Stir casting; Wear; Optimization; SEM

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Paper ID 4030

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Experimental Investigation and Optimization of wear rate in Al6062-SiC-Flyash Metal Matrix Composite

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N.GOVINDU[3]

T.RAVI TEJA[4]

B.KUMAR SRINIVAS RAO[5]

Abstract

Aluminum Metal Matrix composites is a moderately alluring material for vehicle, aviation and other designing application because of its mechanical and tribological properties. To enhance wear protection and mechanical properties has prompted outline and choice of fresher variations of the composite. The present examination manage the investigation of wear conduct of Al6062-SiC-flyash MMCs for shifting fortification substance, connected load, sliding rate, and separation. Aluminum MMCs fortified with three diverse level of support 3, 6, 9% wt. SiC and 0.5, 0.75, 1.00% wt. flyash arranged by mix throwing strategy. Wear test was performed by utilizing "stick on circle" mechanical assembly. An arrangement of analysis in view of L27 Taguchi orthogonal cluster is utilized to obtain the wear information. An examination of fluctuation is utilized to explore the impact of four controlling p sliding separation on dry sliding wear of the composites. It is watched that SiC content, sliding velocity and typical load essentially influence the dry sliding wear. The ideal combina least wear. The microstructure investigation of worn surface demonstrates nature of wear to be for the most part grating.

Keywords:
Metal matrix composites;
Al-SiC-flyash alloy;
Stir casting;
Wear;
Optimization;
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Introduction

Metal Matrix Composites (MMCs) have emerged as a class of material capable of advanced properties, aerospace automotive, electronic, thermal management and wear application. The MMCs have many advantages over the conventional metal including higher specific modules, higher strength to weight ratio, better properties at elevated temperature, and lower coefficients of thermal expansion and better wear resistance. Aluminium composites are widely employed in the aerospace industry, automotive application and structural application. In the present study Al6062 composites were prepared by stir casting method reinforced with different weight % of SiC (3, 6 and 9% wt.) and flyash (0.5, 0.75 and 1.00% wt) as a reinforcement material.

Wear behavior of Al-Mg-Cu alloy reinforced with SiC particle were studied by Hassan [1] the comparative study of alloy and alloy reinforced with SiC suggested that wear resistance property of the alloy increased considerably with addition of SiC particle. Another similar study of wear behavior was performed by Kwok and Lim [2]. The composites were prepared by three different powder metallurgy techniques using different matrix metal, reinforcement weight fraction and reinforcement particle size [2]. In this work the wear studies were conducted at varying reinforcement, load, speed and distance. The wear rates are found to decrease with increasing reinforcement wt. %.

Taguchi Method

Taguchi strategy is one of the essential devices in light of performing assessment or investigations to test the affectability of an arrangement of reaction factors (autonomous or factors) by considering tests in "orthogonal exhibit" with the intend to accomplish the ideal setting of the control parameters. Orthogonal clusters give a best arrangement of very much adjusted (least) tests. Exhibit shows the quantity of line and segments it has, and furthermore the quantity of level in each of the sections. The quantity of line of an orthogonal exhibit speaks to the essential number of analyses. There are five fundamental stage, connected in Taguchi tests outline system. Stage 1-Experiment arranging, Phase 2-Design Experiment, Phase 3-Conducting Experiments, Phase 4-Analyzing Results, and Phase 5-Confirming Prediction Results.

The Taguchi technique utilizes a factual measure of performing called S/N proportion, which is logarithmic capacity of wanted yield to fill in as target work for enhancement, help in information examination and the forecast of the ideal outcomes. There are three sorts of flag to-clamor proportion of regular enthusiasm for enhancement of Static Problem; Smaller-the-better, Larger-the-better, ostensible the-best. They figure for motion to-clamor proportion are outlined so an experimenter can simply choose the biggest factor level setting to advance the quality normal for an examination. For the minimization of wear Larger-the-Better should have been utilized [3].

Paper ID 4034

Analysis Of Process Capability About Turning Center

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Abstract

Analysis of process capability convinces that processes need to be according to manufacturing industry specification, while reducing the process variation and improving the product quality characteristic. The aim is to verify the performance of the process and also the ability of the machine to achieve within the tolerance limit specified. 100 sample shafts were turned on the turning center with subgroup size being ten. The performance needs to be within the control limits and also capable of meeting up to specification. Further, the capability index (Cp) measured in this case is greater than 1. Also, the machine capability (Cpk) for this manufacturing application is also greater than 1, thus making the process capable.

Keywords: Turning center; Control charts; Process capability; Machine ability.

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Paper ID 4035

Effect of Fiber Orientation on Strength of a Cross-Ply Composite Plate using ANSYS

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Abstract

The objective of this paper is to Superior mechanical properties of composite materials such as high stiffness and strength to weight ratios, corrosive resistance and low coefficients of thermal expansion caused it to be used increasingly in many areas of technology including marine, aerospace, automotive and others. To take advantage of full potential of composite materials, accurate models and design method are required the most common structural elements are plates and shells. Laminated composites are one of the classifications of the composites which are used in structural elements like leaf springs, automobile drive shafts, and gears, and axles. An accurate understanding of their structural behaviour is required such as the deflections, the through thickness distributions of stresses and strains, the large deflection behaviour and, of extreme importance for obtaining strong, reliable multi-layered structures. In general, the fibres in the composite materials can be of unidirectional and bidirectional. Here the focus is on unidirectional fibre composite materials. The fibres can be oriented in many ways i.e. 0,10, 20,30,45,90 degrees etc., but the strength in each type will be different. Using the load-deflection graphs, the maximum load and flexural strength are calculated. The attention is to note the variation of flexural strength of a composite material with the variation of fibre orientation and comparing the results with practical results. Here the modelling of composite material with different fibre orientation is done in Ansys software and simulation is done on them and those results are compared with practical results.

Keywords: Flexural strength; Laminated; Fibre reinforced; Glass fibre; Graphite fibre; Unidirectional; Bidirectional; Corrosion resistance.

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* * *

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Paper ID 4001

Design and Methodology of Line Follower Automated Guided Vehicle for Material Handling

SK.Hyderali, Y. Siva Prasad, CH. Naveen Kumar, B. Prasad, V. Dhanunjaya Rao, L. Sai Kumar
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Abstract

Advancement of automated/computerized guided vehicle assumes a noteworthy part in designing businesses to enhance the material taking care of procedure for late year. In this paper, it is focus on the design and different methodology of line follower automated guided vehicle (AGV) systems. This paper provides an overview on line follower AGV discusses recent technological developments. The essential components of line follower robot and their modification are described in this paper.

Keywords: Material handling; Automation; Line follower agv; AGV.

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Paper ID 4031

Fabrication and Experimental Investigation on Mechanical Properties of Aluminium Metal Matrix Composite by using Stir Casting

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Abstract

From the last two decades there is a rapid increase in utilisation of Aluminium Alloy, particularly in Automobile, Aerospace industries, due to its low density, low weight and high strength. Aluminium metal matrix composite has received extensive attention for practical as well as fundamental reasons. Aluminium Alloy & Aluminium based metal matrix composites (MMC) have found many applications in the manufacturing of various components. The present work is focused on Fabrication & Investigation of Mechanical properties of Al6061, Graphite, Fly ash metal matrix composite. These materials are identified in the process of making high strength Aluminium Alloy. Stir casting process is employed for the fabrication of metal matrix composite. In our project the composition of MMC is varied for graphite (3%, 6%) & fly ash (6%, 3%) by keeping 91% by Al 6061. Fabricated samples are tested for Mechanical properties. The Tests includes Tensile test, Hardness test, and Wear test as per ASTM standards. Microstructures of the samples were tested by using metallurgical microscope & compare the microstructure for each.

Keywords: Composite materials; Mechanical properties; Metal matrix composites; Stir casting; Micro sturcture.

Paper ID 4038

Design and Analysis of Aircraft Wing using Composite Materials for varying Angle of Attack

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Abstract

In present scenario, the design of Aircraft became challenging due to problem that are encountered engineering practice. Different analysis are done in the past to develop a better Aircraft. However, th are many parameters that influences the Aircraft design are drag, lift, static and dynamic pressur winglets design, angle of attack, vortex developed, gravity and angle of flight. Hence, the developme in actual practice are not efficient and prone to accidents. In this context, an aircraft wing is developed changing the material for finding better material suitable. The 3D model of NACA 4412 profile is design in CATIA V5 and exported to commercial Ansys 14.5 Software. By applying the boundary conditi similar to the working environment the analysis is carried. The structural analysis of aircraft wing ma of materials such as AA7075 and a composite GLARE (glass-reinforced aluminum laminate) is do under steady inertia loading. Modal analysis also done for defining the natural frequency and mo shape of continuous structures. Further, by varying the angle of attack (AOA) from 00 to 80 for an in velocity of 10m/s the static and dynamic pressure of aerofoil and pressure coefficients are determined using Computational Fluid Dynamics (CFD). The pressure based steady state solver with K-ε turbulen scheme is used for determining the pressures and their coefficients. The results of the analysis are stat in the literature.

Keywords: Air craft wing; Composite materials; Angle of attack; Computational Fluid Dynamics; Glass reinforced composites.

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Design and Analysis of Aircraft Wing using Composite Materials for varying Angle of Attack

A S BhanuPrasanna*
C V Gopinath**
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Paper ID 4039

SHAURIDUT- The solar car of BITS Vizag

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Email: Krantu 6191@gmail.com

Abstract

In urban communities of India one of the real medium of transportation is auto rickshaws, whicl delivering a colossal measure of air contamination and also nursery gasses like CO2. Fuel, whic utilized is a non-inexhaustible source and furthermore which costs, one of the front runners within vicinity of renewable energy resources nowadays is sun strength. a solid source like sun oriented vit which is accessible in bounty in a nation like India. Received SOLAR ENERGY as the extra sou notwithstanding the regular IC Motors. Utilizing the Photovoltaic panels, controllers, and brushless motor setup to change over the light, as an electric vitality which is sustained to the DC engine to acc

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HEPSIBA SEELI*
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Abstract

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The paper represents how the charge produced by a variety of sun-powered boards are gotten and its stream all through a battery pack is to be controlled utilizing a microcontroller based charge controller to guarantee effective putting away of charge in a battery pack. The putaway vitality would be unveiled to a DC engine which would run the auto. The design of an engine controller to control the auto's speed and forward/turn around bearing of movement appears. The mechanical development from scratch of the chassis along with all crucial mechanical frameworks of Shauridut solar car is illustrated.

Keywords:

Solar car, Solar Energy, Photovoltaic panels, Brushless DC motor, CONTROLLER, Batteries,

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Paper ID 4040

Numerical And Experimental Analysis Of Thermal Performance In A Parallel Flow And Counter Flow Double Pipe Heat Exchange

Hepsiba Seeli*, B. Bhanu Chander Rao**, K Laxmana Rao***, Rajesh Romala****
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Abstract

Heat transfer is taken into consideration as transfer of thermal energy from bodily frame to another. It is the most essential parameter to be measured because the overall performance and efficiency of the double pipe heat exchanger. By the use of CFD simulation software, it reduces the time and operation cost in comparison to experimental calculations, The goal of this paper is to evaluate the influence of the Double pipe heat exchanger to get better heat transfer. The purpose of this examine is to apply CFD software program and Experimental setup to analyze the Temperature drop, Pressure drop and Friction factors, by varying under different inlet conditions like Temperature and Flow rate as a function of both inlet velocity and temperature variations and converting heat exchanger tube material properties like copper and aluminium. In this experiment, heat transfer from hot fluid to cold fluid with the aid of Double pipe heat exchanger is experimentally by using different organic fluids like Benzene, Glycol, Transformer oil, Acetone and Water to get better heat transfer, the identical thing is established in CFD analysis. The test is accomplished at Laminar flow beneath different flow preparations like Parallel flow and Counterflow. The test is done on double pipe heat exchanger under various working fluids with different operating conditions, it is predicted that Counterflow exhibits better heat transfer than Parallel flow.

Keywords: Heat transfer, Double pipe heat exchanger, CFD simulation software, Organic fluids, Copper, Aluminium, Laminar flow, Turbulent flow, Parallel flow and Counterflow

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Paper ID 4041

Experimental Data of Calophyllum Inophyllum Bio Diesel Effect by Varying Compression Ratio and Injection Pressure on Emissions

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Abstract

Degrading fossil fuel resources have thrown world into chaos, besides air pollution and green hou gases are a major concern. The fossil fuel shortage in near future is inevitable. This situation triggered tl awareness to find alternative and sustainable energy resources. Biodiesel is a better solution as its properti are close to petrodeisel. In this paper, Biodiesel is produced from Calophyllum inophyllum oil by tv stage transesterification process using acid and base catalysts. Acid esterification is done by mixi 1%(by wt. of oil)of 99% pure anhydrous sulphuric acid in 1000 ml of oil with oil to methanol ratio 6 and maintained at a temperature of 55R C for 120 minutes. This reduces the acid value of the oil. Nc the oil is subjected to base catalyst. 1% of KOH (by wt.of oil) is added to a mixture of oil to methan ratio4:1. The mixture is stirred for two hours at 60R C. The yield is about 89%. Due to high viscosity is difficult for regular engines to inject biodiesel into combustion chamber. This results in improp mixing of fuel and reduces ignition quality. In this paper the fuel injection pressure has been increas and compression ratio has been varied and tested under different operating conditions. The increase injection pressure resulted in better atomization of fuel and better mixing with the air in combusti chamber. The increase in compression ratio resulted in higher compression pressure which ensures compl combustion of fuel in the chamber. These modifications results in reduction of emissions and smoke to about 10% compared to normal engines running on biodiesel.

Keywords: Biodiesel, Transesterification, VCR Engine, Injection pressure, compression ratio, Eng emission.

Paper ID 4041

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Fabrication and Experimental Investigation on Mechanical Properties of Aluminium Metal Matrix Composite by using Stir Casting

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Abstract

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Paper ID 4033

Reduction Of Bio Diesel Emissions By Injection Advance In VCR Engine

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Abstract

Biodiesel in present trend is important due to depletion of non-renewable resources. The shortage of fossil fuels brings usage of bio diesel into limelight. Therefore, usage of bio diesel is preferred than usage of conventional diesel as the properties of both are almost same. A lot of research has undergone on bio diesel and doesn't show any significant variation in emissions compared to conventional diesel. Due to high emissions of harmful gases global warming increases which is not preferable. In this paper Flax seed is used for bio diesel preparation which is abundantly available in mountainous areas and cooler regions. Flax seed commonly known as lin seed has binomial name Linum usitatissium. In this process the oil is transesterified using methanol to produce methyl esters in presence of base catalyst. Methanol is mixed with flax seed oil in the ratio 1:5 and 0.5 to 5% base catalyst is added. This mixture is constantly stirred at 55R C. Due to high viscosity it is difficult for biodiesel to mix with air in the combustion chamber and results in increase of emissions . As the fuel injection is advanced by change in crank angle up to certain extent in a VCR engine, the time available for mixing fuel with air is increased ensuring increased mixing of fuel with air, and By increasing the compression ratio the pressure in combustion chamber increases which helps high viscous fuel to mix completely, due to high temperature because of the high pressure in the combustion chamber the combustion characteristics of charge increased results in lowering the engine emissions. The above tests were conducted using conventional diesel, and different proportions of blends of bio diesel and results were compared by varying crank angle and compression ratios.

Keywords: Biodiesel, Transesterification, Crank angle, Emissions

* * *

Paper ID 4033

Reduction Of Bio Diesel Emissions By Injection Advance In VCR Engine

CH Uma Chaithanya*, CH Kodanda Rama Rao**

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Keywords: Biodiesel, Transesterification, Crank angle, Emissions

* * *

Paper ID 4045

Extracting oxygen from exhaust gas

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B.Shiv Ganesh, Assistant Professor, Mechanical Department, Baba institute of technology and science. Email: shiv.ganesh364@gmail.com

Abstract

As a engineer our first goal is to create a friendly and safe environment to humans. Most precious thi in the world is life. In race with life every second is precious. Lots of people losses there life in accider due to lack of oxygen. To overcome this we would like to generate oxygen from exhaust gases whi were released from automobiles. In this we take any one type of gas from exhaust gases in this by doi chemical reactions we separate oxygen from that exhaust gas which is further used in medical emergen cases. In this we create a extra chamber on the top of the catalytic converter in which oxygen is separat and stored by where oxygen is supplied in emergency conditions. In medical emergency we can sup oxygen to the patient from our bike or car till ambulance reaches to the spot. By this we can save patient life.

Keywords: zeolite, catalytic converter.

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Paper ID :4046

Experimental studies on double pipe heat exchanger for heat transfer enhancement by using nanofluids

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Abstract

Heat transfer is one of the most important processes in many industries. Base heat transfer fluid like water normally is having low thermal conductivity. Since thermal conductivity is considered as important factor for rapid cooling and heating application, advanced heat transfer fluid called nano fluid is introduced. Nano particle is low concentration cooling agent which uniformly scattered in nano fluid Suspensions of Nano particles shows better enhancement of stability, heat transfer capabilities and reduction of particle clogging. All researchers tried to increase the heat transfer rate by controlling the parameters like shape, size, clustering, collision, porous layer, melting point of nanoparticle. Previously we were using only refrigerant as working fluid in heat exchanger for cooling purpose. The presence of fluorine and chlorine make the refrigerant non toxic. Hence the whole world attracted towards the Nano fluid. Finally this project is to be carried out through Concentric Double pipe Heat Exchanger with base fluid water and nano particles like MnO_2. Study is made to ascertain the effect of Nano fluids in enhancing the effectiveness of heat exchanger by varying the concentration of nano fluids.

Keywords:Thermal conductivity;Heat transfer;Nano fluid;Double pipe heat exchanger;Nano particels.

* * *

Paper ID 4047

Material Evaluation for Manufacturing Impeller Blade of a Steam Exhauster in Visakhapatnam Steel Plant

C. V. Gopinath, Uppada Venkata Ramana

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Abstract

The Steel Melt Shop (SMS) of Visakhapatnam steel plant is one of the core departments. The steel plant's productivity is entirely dependent on the performance of the steel melt shop. The hot metal from 'Blast Furnace' is converted into semi-finished product (Blooms) in the 'Tundish Machines'. For proper solidification, water is sprinkled over the liquid steel through the mould. This process generates steam which has to be exhausted to the atmosphere using steam exhauster.

The steam exhauster is placed in the connecting pipe which connects the bunker where solidification takes place and the chimney. The impeller blades of the steam exhauster are of complex shape and are made of AISI 904 L materials. These are subjected to various thrust forces and vibrations which cause various damages.

The present project is to study these damages and analyze the impeller blades due to the static loads. The modeling of the impeller blade assembly is carried through CATIA V12 R18 using "Part Design" and

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The present project is to study these damages and analyze the impeller blades due to the static loads. The modeling of the impeller blade assembly is carried through CATIA V12 R18 using "Part Design" and

"Wire frame and surface design" module. The solid model of the blade is then imported to ANSYS software where static analysis is carried out. Various pressure loads and different materials and design optimization are considered to analyze the blade in the project.

Keywords: Steam exhauster; Impeller; AISI 904L; Part Design; wire frame and Surface Design;

* * *

Paper ID 4048

Optimum Material Evaluation for Manufacturing L-Angle of 3-Wheeler

C. V. Gopinath* & Chitturi Tejaswi*

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Abstract

In the present scenario of automobile sector, auto rickshaws (3-wheelers) are the most sought after public transport among the urban and rural poor in India. Majority of people depend on these vehicles after public transport systems like Road Transport Corporation. Among these 3-wheelers, 7-seaters are the most popular and a lot of developments are seen in the past few years. MLR Motors Pvt Ltd, Hyderabad is a 3- wheeler manufacturer which is trying to mark its presence in the market. "TEJA" The vehicle manufactured by the company has a seating capacity of seven persons including the driver. The vehicle has a worm and roller type steering gear mechanism unlike other conventional auto rickshaws fitted with handle bars. Manufacturing of an auto rickshaw involves assemblies of several major components. L Angle of the auto rickshaw is one such major component. It connects the floor and front panel and plays a vital role in design and development of the vehicle. The project work deals with modeling and analysis of the L-angle, subjected to structural loads. The analysis is carried out to optimize the L-angle by analyzing the behavior of the L-Angle for varying thicknesses, different materials and modifications in design. The modeling of L-angle is first carried out in CATIA V5 R18 using Part Drawing and Surface Modeling module. The analysis is done using ANSYS 11.0 by importing the solid model from CATIA in IGES format. During the static and modal analysis the model is subjected to various loads, the results are derived and the optimized L-angle is proposed.

Keywords: Surface Modeling; Static Analysis; Optimization; Automotive Design;

* * *

Paper ID 4049

Buckling Analysis on Thin Films Mechanical-Thermal Coupled-Field Using Theoretical Calculations

Dasari Gowri Sankar* & Dharmala Venkata Padmaja**

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Abstract

Devices with feature size on the order of one micrometer have found wide spread applications in science and engineering. MEMS, is a rapidly growing technology for the fabrication of miniature devices which

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Buckling Analysis on Thin FilmsMechanical-Thermal Coupled-Field Using Theorotical Calculations

Dasari Gowri Sankar*
Dharmala Venkata Padmaja**

Abstract

Devices with feature size on the order of one micrometer have found wide spread applications in science and engineering. MEMS, is a rapidly growing technology for the fabrication of miniature devices which provides a way to integrate mechanical, fluidic, optical, and electronic functionality on very small devices, ranging from 0.1 microns to one millimeter. Recently, it has been proposed that regular patterns can be generated through the mechanical buckling of a thin film. The process of buckling of thin compressed films deposited on polymethylmethacrylate (PMMA) Vs Structral Steel compared under mechanical and thermal loadings has been investigated utilizing an optical microscope. Particularly, thermal cycling analysis on thin film/substrate system under compression has been characterized to discuss the thermal property of PMMA and Structral substrate validate through Therotical Calculations.

Keywords:
Mechanical Buckling
Thin Flims
Miniature Device

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1. Introduction:

Film/substrate structure in information science occupies an important position, for example, data storage and processing systems on integrated circuits contain a large number of conductive, semi-conductive and insulating films, the magnetic films which play a key role in the disk storage systems, etc.

However, the above-mentioned thin film/substrate system withstand a variety of load at work (such as cutting tools, anti-corrosion coating), thermal stress caused by the heat (such as the micro-chippackaging coating), especially the residual stress in the film, either the thermal mismatch produced in the high-temperature deposition process and the subsequent cooling process, or the internal stress caused by lattice mismatch. Because delamination and buckling is the main failure modes of these devices, so the investigation of buckling is of great significance for its life prediction.

Thin-film buckling (buckle) generally considered refers to a failure mode caused by the compressive force on the film material, characterized by the vertical displacement perpendicular to load direction. Buckling based on specific patterns can be divided into the straight-sided wrinkle; the circular blister or bubble; the telephone cord buckle; The expansion model of straight-sided wrinkle is recognized in the steady-state expansion conditions that the wrinkle expands along the curved front of the oval; while the linear edge formatted subsequently do not expand. The cross-section of straight-sided wrinkle shows the shape of cosine curve. The initial expansion maintains round, but with the increase of residual stress, the bubble mutated into lobe. Some circular buckles have small straight-

Paper ID 4052

Fabrication of Rocker Bogie Suspension System

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eywords: Rocker; Bogie

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Abstract

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Keywords:

Rocker;
Bogie;

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1. Introduction

The rocker bogie suspension framework, which uncommonly intended for space investigation vehicles have profound history implanted in its improvement. The expression "Rocker" portrays the shaking part of the bigger connections exhibits each side of the suspension framework and adjust the bogie as these rockers are associated with each other and the vehicle case through a specifically altered differential.

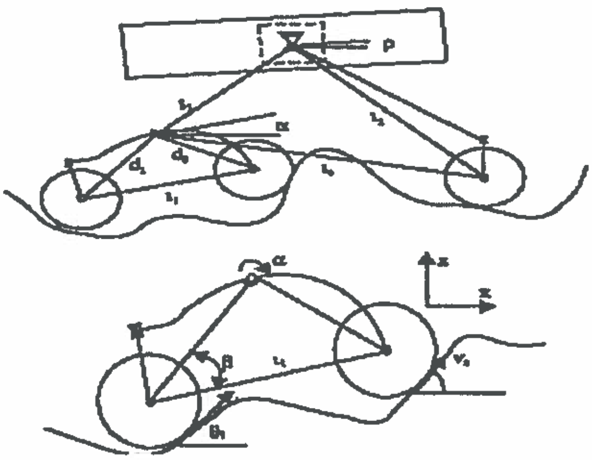
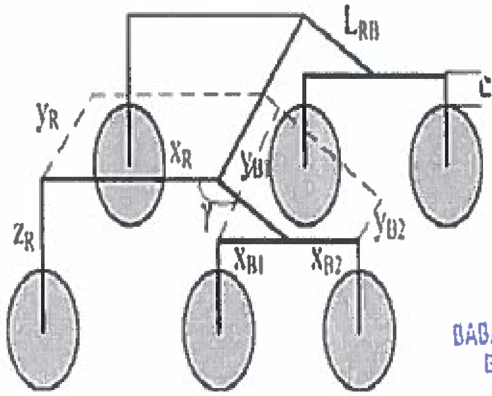


Fig. 1 Schematic diagram of rocker bogie suspension system

As agreement with the movement to keep up focal point of gravity of the whole vehicle, when one of the rocker moves upward, alternate goes down. The case assumes fundamental part to keep up the normal pitch point of the two rockers by enabling the two rockers to move according to the circumstance. According to the intense plan, one of the finishes of a rocker is fitted with a drive haggle opposite end is turned to a bogie which gives required movement and level of opportunity.



Paper ID 4055

Design and fabrication of Three Axes Pneumatic Trailer for shipping Goods

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Abstract

The objective of this paper is to evaluate the difficulty in unloading the materials. In this regard, seve automobiles garages have revealed the facts that, most difficult methods were adopted in unloading materials from the trailer. The trailer can load or unload the material in only one single direction. Moreov It is difficult to unload the materials in small compact streets and small roads. Now our project l mainly concentrated to rectifythe loading and unloading of the trailer in all the three sides very eas with suitable arrangement of design. The vehicles can be loaded from the trailer in three axes with application of any impact force by activating the direction control valve. Compressed air is used controlling the valve. The ram of the pneumatic cylinder acts as a lifter of trailer cabin. The automob engine drive is coupled to the compressor engine, so that it stores the compressed air when the vehi runs. This compressed air is used to activate the pneumatic cylinder, when the valve is activated.

Keywords: Trailers, Control Valve, Pneumatic Cylinder, Compressed air

* * *

Paper ID 4056

Dynamic Analysis On R31 Steam Turbine Blade

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Abstract

The Steam turbine blades are subjected to in-plane load because of fluid or aerodynamic pressures so tl blades are subjected to high dynamic loadings. It is necessary to do vibration analysis to know the dynam characteristics of the material as they are working at high speeds. The X20cr13 steel material R31 turbir blades are used in 7.5MW BBC as turbine rotor in power plants. Flow of the super-heated steam in steai turbines causes failure of blades due to high temperature working condition, high rotational speeds an corrosion of blade due to the purity of steam. The existing X20 turbine blade is replaced with the composi material without the increase in the stresses. Two types of composites are taken with a matrix an reinforcements. One is glass-ceramic matrix systems reinforced with silicon carbide and silicon carbid reinforced with the aluminum matrix. Aluminum silicon carbide compare the mode of vibrations for th existing and newly design composite plates by using CATIA V5 R21 and ANSYS 15.0 software's. Secondl the existing blade made of X20Cr13 is replaced with the metal matrix composite aluminum silicon carbide Metal matrix Composites possesses high-temperature capability, high thermal conductivity, low CTE and high specific stiffness and strength. Addition of SiC into the aluminium matrix produces a resultin

composite with the low density of aluminium but a higher modulus similar to steel. When the existing X20 material turbine blade is replaced with the composite material AlSiC MMC and Pyrosic, the von mises stresses are very much reduced. The life of the blade is also increased by using composite materials.

Keywords: ANSYS, CATIA V5, Stress Analysis, Composite, Fatigue, Modal Analysis

* * *

Paper ID 4057

Design and Analysis of Single Point Cutting Tool by Varying rake angle

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Abstract

In the engineering industry Metal cutting process forms the basis and is involved indirectly or directly in the manufacture of every product of our modern civilization. The study of metal cutting is very important and knowledge in the fundamentals of machining of materials is essential. Theory of metal cutting will help to develop a scientific approach in solving problems encountered in machining. In this project "Design and Analysis of Single Point Cutting Tool by varying rake angle", by varying rake angles the metal cutting theory is studied and this have been extended to tool-geometry, materials used and its properties, working conditions and its characteristics, effects of variable parameters like feed, cutting speed, and depth of cut during the machining process. This project includes modeling of single point cutting tool with commercial angles in CATIAV5R20 and then the analysis is done in ANSYS15.0. After post processing the results of analysis, we modified the geometrical design and designed various tools with different back rake angles, analyzed them individually.

Keywords: cutting tool; cutting speed; feed; depth; Catia; Ansys.

* * *

Paper ID 4058

Design And Farication Of Industrial Parameter Monitoring And Control Robot

CH Uma Chaithanya, K Sri Ram Murthy, P Siva Vasu, D Rama Krushna, K Yuva Kishore, K P Bala Krishna
CH Uma Chaithanya, Assistant Professor, Department Of Mecanical Engg, BITS-Vizag,India Email: uma.nitw@gmail.com

Abstract

Now-a-days the accidents in the coal mine industries have increased. Even if any explosion occurs it can't be easily known to the worker and it may cause accidents. So in order to avoid this, a robot has been designed and this robot is allowed to monitor the ambient situations inside the coal mine industry. Some of the environmental parameters such as methane leakage, temperature, oxygen are sensed by using the high end sensors and the sensed data are transmitted for monitoring the status of the coal mine and to control the robot movement. If the temperature exceeds a threshold, the cooling fan is automatically set

to ON and if any gas leakage is detected the workers are given alert through a buzzer. By this the human intervention can be avoided inside the industry and the accidents can be prevented.

Keywords: DC Motors; Sensors; Buzzer;Battery;

* * *

Paper ID 4059

Analysis of Bonded Tubular Single Lap Joints subjected to varying Torsion at constant Pressure

Vikas Ranjan, Dr. R.R Das

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Dr. R.R Das,Professor, department of mechanical engineering, ISM Dhanbad, India. das.rr.mme@ismdhanbad.ac.in

Abstract

The present research deals with Finite Element Method (FEM) based stress and failure analyses of different types of adhesive bonded tubular joint configurations, viz. Tubular Single Lap Joint (TSLJ) and Tubular Socket Joints (TSJ). The bonded joints have been made with laminated FRP composites and have been subjected to different loading conditions viz. circumferential, axial, and torsion. Suitable APDL codes have been developed using ANSYS 14.0 which is capable of capturing the stresses three dimensionally within the joint region. The developed FE model has been validated with respect to analytical results available in literature and found to be in good agreement. Three-dimensional stresses within the joint and different adherend-adhesive interfaces which in turn play key roles in initiating the failures have been studied in details. Locations prone to adhesion and cohesion failures in the overlap or coupling regions have been identified using Tsai-Wü and Parabolic yield criteria based failure indices. Suitable joint parameters and ply-orientations for the laminated FRP composite joints have been suggested in order to improve the performance and resistance of the bonded tubular structures against failures.

Keywords: Finite Element Method (FEM), Tubular Single Lap Joint (TSLJ), Tubular Socket Joints (TSJ) Fibre Reinforced Polymer (FRP).

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PRINCIPAL
BABA INSTITUTE OF TECHNOLOGY AND SCIENCES
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VISAKHAPATNAM - 530 048

Paper ID 4061

Engraving letters by CNC Milling

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Students, Department of Mechanical Engineering, Baba Institute of Technology and Sciences, Visakhapatnam.

Abstract

CNC is a machine controlled by a computer with command data code numbers, letters and symbols in accordance to conventional ISO. Computer Numerical Control machine tool systems work similar then the CNC machine tools more accurate, more precise, more flexible and appropriate for the manufacturing. This project describes a programming of letters by CNC milling machine. This project applied working standards of CNC milling device, in which it can flow in 3 axes in particular X, Y and Z. We design to assist engrave the letters of BITS which requires a excessive degree of complexity and might reduce operator intervention throughout machine process. For this project, Master CAM software was used to generate the G & M-codes for milling cutting. Master cam Mill is the next generation popular program, delivering the most comprehensive milling package with powerful new tool paths and techniques. This paper highlights the CNC G-Code Text Engraving on an aluminum work piece.

Keywords: Computer Numerical Control, Engraving, G & M-codes, CNC milling, 3- axes, Master CAM

* * *

PAPER ID: 5004

Design of an Aperture (Pyramidal horn) antenna for wide bandapplications using CST

Ms. G.Kanchana Mr. I.B.A.Sarath

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Abstract

The aperture antenna is also known as pyramidal horn antenna that operates in microwave band i.e., 300 MHz to 300 GHz. They are extensively used in the fields of T.V broadcasting, microwave devices and satellite communication. Since horn antennas do not have any resonant elements they operate at wide range of frequencies and have a wide bandwidth. They are also used as high gain devices in phased arrays and as a feeder for reflector and lens antennas in satellite communication. The designed aperture antenna is functional for each UHF band applications and here it is having gain of 5dB operating at 2.8 GHz frequency. The performance parameters like Directivity, impedance, Efficiency, s-parameters are evaluated using CST.

Keywords: Aperture (Pyramidal horn) antenna, Gain, Bandwidth, Beam width, Radiation Pattern, Computer Simulation Technology

* * *

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PAPER ID: 5011

Embedded Based Efficient Railway Track Crack Detection System Using Zigbee Technology And Gps

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Associate Professor, Department of ECE, BITS, Visakhapatnam, A.P

Abstract

Today's most of the railroad investigation is manually conducted by track examiners. Practically, it is not easy to investigate the thousands of railway track by trained human examiners. Hence it takes too much time to inspect the defected railway tracks and then inform to the railway authority people. In order to avoid delay and improve the accuracy, we have proposed structure that will automatically investigate the railway cracks and obstacles by using inspection robot, then the system also concludes automatic gate control system. The cracks and obstacles are detected by using sensors in robot, which are then informed to the train driver through the GSM using radio frequency signals. In this proposed system the train can be controlled without driver control when the problem is identified. These detection and controlling methods can be used to prevent the railway accidents.

Keywords: Railway track inspection, robot control, train control, gate control, Ultrasonic sensor, Infra-red sensor, GSM,GPS,RF transmitter and receiver.

* * *

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* * *

Automatic Dialing To Any Phone Using I2c Protocol On Detecting Burglary

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B. Siva Prasad , Associate Professor , BABA Institute of Technology and Sciences ,Visakhapatnam.

Abstract

The main objective of this paper is to intimate the concerned authorities about an unauthorized access of secured areas such as museums, residential houses, banks etc. As crime rate is on the rise and burglars are getting smarter, the security system for banks, shops and houses needs to be full proof. This proposed system ensures that at any moment if any unauthorized person tries to open the bank locker; a mobile number will be dialed through the GSM modem connected to it.

Keywords: GSM Modem, EEPROM, Fingerprint scanner, AT89S52, I2C Protocol.

* * *

PAPER ID: 5016

DESIGN OF RNS MONTGOMERY MULTIPLICATION ARCHITECTURE

N.Nandini1, B. V.R.Gowri2, B.Siva Prasad3

N.Nandini. PG Scholar, Baba Institute of Technology and Sciences,Visakhapatnam. Email: nnadikanti11@gmail.com.

Abstract

A design methodology for incorporating Residue Number System(RNS)/polynomial Residue Number (PRNS) in Montgomery modular multiplication in r respectively, as well as a VLSI architecture of a dual -field residue arithmetic Montgomery multiplier are presented in this paper. As analysis of input/output conversions to/from residue representations, along with the proposed residue Montgomery multiplication algorithm, revels common multiply-accumulate data paths both between the converters and between the two residue representations. A versatile architecture is derived that supports all operations of Montgomery multiplication in and, input/output conversions, Chinese Reminder Theorem(CRT)/Mixed Radix Conversion(MRC) for integers and polynomials, dual -field modular exponentiation and inversion and same hardware. Detailed comparisons with state-of-the-art implementations prove the potential of residue arithmetic exploitation in dual-field modular multiplication. We present a low latency systolic Montgomery multiplier over GF(2m) based on irreducible pentanomials. For the parallel processing the decompose of the multiplication into a number of independent units by using an efficient algorithm. "pre-computed addition" technique is introduced to further reduce the latency. Comparing with other existing methods

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Design of RNS Montgomery Multiplication Architecture

N.Nandini*

B.V.R.Gowri**

B.Siva Prasad***

Keywords:

Xilinx;RNS Value;

Montgomery;

Mixed radix conversion.

Abstract

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1.Introduction

Everyday transactions from few years ago required some physical existence have been replaced by electronic applications. Users may now use friendly, fast and safe interfaces to perform easily numerous tasks, varying from money transfers using the web and remote health checks to e-learning and e-commerce. Being

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Implementation of AES Architecture Design In High Level Cryptography

Ch.SantoshiKumari*
P.K.Suresh**
K.Pradeep***

Keywords:

AES,

ENCYPTION,

DECRYPTION,

XILINX12.3i.

Abstract

The selective application of technological and related procedural safeguards is an important responsibility of every Federal organization in giving sufficient security to its electronic data systems. This publication explains a cryptographic algorithm, the Advanced Encryption Standard (AES) which can be utilized by Federal organization systems to save sensitive information. Safety of data during transmission or while in storage may be compulsory to keep up the confidentiality and integrity of the information provide by the data.

The algorithm clearly explains the mathematical process needed to send the data into a cryptographic cipher and also to transform the cipher back to the original position. The advanced Encryption Standard is being made available for use by Federal organizations within the situation of a total security program consisting of physical safety procedures, fine information management practices, and computer system/network access controls. In this the key should be available at the transmitter and receiver simultaneously during communication. In this project both encryption and decryption procedures are executed. The functioning is observed using ISE simulator and the synthesis is carried out using XILINX ISE 12.3i.

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P.G.Scholar,

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1. Introduction

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*** English Language Specialist, Oller Center, Carriage House, 2nd Floor, California, USA

PAPER ID: 5019

Analysis Of "bloom Filter" Using Cam Based Algorithm

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Email: k.pradeep423@gmail.com, rmsh_ece@yahoo.co.in

Abstract

Content-addressable memory (CAM) employing a new algorithm for associativity between the input tag and the corresponding address of the output data. The proposed architecture is based on a recently developed sparse clustered network using binary connections that on-average eliminates most of the parallel comparisons per- formed during a search. Given an input tag, the proposed architecture computes a few possibilities for the location of the matched tag and performs the comparisons on them to locate a single valid match.Bloom filters (BFs) provide a fast and efficient way to check whether a given element belongs to a set. Reliability is becoming a challenge for advanced electronic circuits as the number of errors due to manufacturing variations, radiation, and reduced noise margins increase as technology scales. In this brief, it is shown that BFs can be used to detect and correct errors in their associated data set. This allows a synergetic reuse of existing BFs to also detect and correct errors. The proposed scheme Content-addressable memory (CAM) is a special type of computer memory used in certain very-high-speed searching applications. It is also known as associative memory, associative storage, or associative array, although the last term is more often used for a programming data structure. It compares input search data (tag) against a table of stored data, and returns the address of matching data (or in the case of associative memory, the matching data). Several custom computers, like the Goodyear STARAN, were built to implement CAM, and were designated associative computers.

Keywords: Bloom filters (BFs), error correction, Soft errors.

PAPER ID: 5021

Low Power High Speed Low Latency Bi Rotational Cordic Architecture

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Abstract

CORDIC is an acronym for Coordinate Rotation Digital Computer. The CORDIC calculation is a d… figuring approach capacity of developing distinctive essential capacities with an appropriate move an… add technique Used to assess a lot of capacities. It has been utilized for a long time for proficient usage… vector revolution operations in equipment. It is executed simply by table look-into, move, and expansi… operations. Revolution of vectors through settled and known points has numerous applications… livelinesss, apply autonomy, amusements, PC illustrations and advanced flag preparing. In this paper… display upgraded strategy for moving by using a substitute arrangement by extending the no. of bar… shifters with growing pre moving methodology and Fault Tolerance in Bi Rotational CORDIC circ… higher rate of precision in settled and known turns. The adjustment in the settled point Rotation lesse… the district and Complexity in the application. From the basic building of cordic a settled point rota… executed by vector turn. the upheaval of vectors uncontrolled by the circuit till all turns are done it… occurs significant system get and whimsical plots for convincing operation of known edges in this pap… edge update ,Quadrant change and get correction is executed.

Key words: CORDIC, Quadrant amendment, vectoring and rotation modes.

* * *

PAPER ID: 5022

Implementation of AES Architecture Design in High Level Cryptography

Ch.Santoshi Kumari, P.G.Scholar, Chintasantoshi701@gmail.com
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K.Pradeep, Associate Professors, pradeepbitsvizag@gmail.com

Abstract

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The algorithm clearly explains the mathematical process needed to send the data into a cryptographic cipher and also to transform the cipher back to the original position. The advanced Encryption Standard is being made available for use by Federal organizations within the situation of a total security program consisting of physical safety procedures, fine information management practices, and computer system/network access controls. In this the key should be available at the transmitter and receiver simultaneously during communication. In this project both encryption and decryption procedures are executed. The functioning is observed using ISE simulator and the synthesis is carried out using XILINX ISE 12.3i.

Keywords:

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PAPER ID: 5025

Design Of High Speed Low Power Parallel Adder Using Bec Technique

Kala Priya.K1, Rajani Chandra CH2
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Assistant Professor, Department of ECE, BITS, Visakhapatnam, India2

Abstract

Addition is a fundamental operation for any digital system, digital signal processing or control system. / fast and accurate operation of a digital system is greatly influenced by the performance of the resider

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DESIGN OF HIGH SPEED LOW POWER PARALLEL ADDER USING BEC TECHNIQUE

Kala Priya.K*

Rajani Chandra CH**

Abstract (10pt)

Addition is a fundamental operation for any digital system, digital signal processing or control system. A fast and accurate operation of a digital system is greatly influenced by the performance of the resident adders.Adders are also very important component in digital systems because of their extensive use in other basic digital operations such as subtraction, multiplication and division. Hence, improving performance of the digital adder would greatly advance the execution of binary operations inside a circuit compromised of such blocks. The performance of a digital circuit block is gauged by analysingits power dissipation, layout area and its operating speed. As a result, the need for faster and efficient Adders in computers has been a topic of interest over decades.

Binary adders are one of the most essential logic elements within a digital system. In addition, binary adders are also helpful in units other than Arithmetic Logic Units (ALU), such as multipliers, dividers and memory addressing. Therefore, binary addition is essential that any improvement in binary addition can result in a performance boost for any computing system and helps to improve the performance of the entire system.The major problem for binary addition is the carry chain. As the width of the input operand increases, the length of the carry chain increases. The binary adder is the critical element in most digital circuit designs including digital signal processors (DSP) and microprocessor data path units. As such, extensive research continues to be focused on improving the power delay performance of the adder. The basic idea of the proposed work is using various types of addersand compared to the adder with Binary to Excess 1 converter (BEC) for high speed of addition by using the simulation through Verilog in Xilinx ISE.

Keywords:

Adders,Binary adders, Parallel-prefix adders, Verilog, Xilinx

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- Rajani Chandra CH, Assistant Professor, Department of ECE, BITS, Visakhapatnam, India.

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Keywords: Adders, Binary adders, Parallel-prefix adders, Verilog, Xilinx

* * *

PAPER ID: 5026

Optimal Capacitor Placement using Fuzzy Logic and Genetic Algorithms for Radial Distribution Systems

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Abstract

This paper presents a fuzzy logic based approach to determine suitable nodes in a radial distribution system for capacitor placement and genetic algorithms for sizing the capacitors. Voltages and power loss indices of Radial Distribution System(RDS) are modeled by fuzzy membership functions. A Fuzzy Inference System (FIS) containing a set of rules is used to determine the capacitor placement suitability(CPS) of each node in the Radial Distribution System. Capacitors are placed on the nodes with the highest suitability. The installation of power capacitors in RDS improves voltage profile and reduces the real power losses . The capacitor sizing is done using genetic algorithms(GA) for improving the annual cost savings of Radial Distribution System.

Keywords: Radial distribution system, Fuzzy inference system, genetic algorithms, power loss index, Capacitor placement.

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adders.Adders are also very important component in digital systems because of their extensive use in other basic digital operations such as subtraction, multiplication and division. Hence, improving performance of the digital adder would greatly advance the execution of binary operations inside a circuit compromised of such blocks. The performance of a digital circuit block is gauged by analysinglits power dissipation, layout area and its operating speed. As a result, the need for faster and efficient Adders in computers has been a topic of interest over decades.

Binary adders are one of the most essential logic elements within a digital system. In addition, binary adders are also helpful in units other than Arithmetic Logic Units (ALU), such as multipliers, dividers and memory addressing. Therefore, binary addition is essential that any improvement in binary addition can result in a performance boost for any computing system and helps to improve the performance of the entire system.The major problem for binary addition is the carry chain. As the width of the input operand increases, the length of the carry chain increases. The binary adder is the critical element in most digital circuit designs including digital signal processors (DSP) and microprocessor data path units. As such, extensive research continues to be focused on improving the power delay performance of the adder. The basic idea of the proposed work is using various types of addersand compared to the adder with Binary to Excess 1 converter (BEC) for high speed of addition by using the simulation through Verilog in Xilinx ISE.

Keywords: Adders, Binary adders, Parallel-prefix adders, Verilog, Xilinx

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PAPER ID: 5027

Fuzzy Based Sfcl For Limiting Fault Current In Distribution Generation

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Elipilli Anil Kumar2,Assistantprofessor Bits Vizag
Kasi Venkateswara Rao3,Assistant Professor Bits Vizag

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PAPER ID: 5028

Development of a Photovoltaic array in Simulink and its Performance Verification

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Abstract

This paper presents Simulink-based PV module model of various types of Photo-Voltaic(PV) cells which includes a controlled current source and a MATLAB function with the assistance of Simpowersystems Tool box. The PV Cell is the main building block of a PV array for simulating and monitoring the performance. The modeling scheme is done by using some predigested functions in MATLAB. With the proposed model, we can predict the behaviour of a photo voltaic cell under the conditions of both non-uniform and uniform irradiance which is very useful for design engineers to determine the performance of any photovoltaic array in a easier way without performing any tedious iterative numerical calculations. The results are compared with the practical values obtained from the solar panel.

Keywords: PV cell; Modelling;

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PAPER ID: 5029

Power Quality Improvement In Distribution System By Using Dynamic Voltage Restorer

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Abstract

In this paper, another control methodology in light of current mode control for Dynamic Voltage Restorer (DVR) is proposed to relieve the power quality issues in the supply voltage. The DVR is controlled in a roundabout way by controlling the supply current. The reference supply streams are evaluated utilizing the detected load terminal voltages and the dc transport voltage of DVR. The control conspire depends on synchronous reference outline hypothesis (SRFT) for the operation of a capacitor upheld DVR. The control methodology is confirmed through broad reproduction ponders utilizing MATLAB with its .mulink and Power System Blockset (PSB) tool stash to exhibit the made strides execution of DVR.

Keywords: Power Quality; DVR; Custom Power devices; Current control.

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PAPER ID: 5030

AC Line Power Factor Correction of DC Motor Drive employing Boost Converter

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Abstract

With the expanding interest for control from the ac line and more stringent breaking points for control quality, control factor rectification has increased awesome consideration as of late. A variety of circuit topologies and control systems has been delivered for the PFC application. While the intermittent conduction mode (DCM) converters, for example, help and flyback converters are appropriate for low power applications, nonstop conduction mode (CCM) support converters with normal current mode, top current mode or hysteresis control are regularly decided for some medium and high power applications. Consonant pollution and low power factor in charge structures caused by control converters have been of amazing concern. To vanquish these issues a couple of converter topologies using pushed semiconductor contraptions and control designs have been proposed. This examination is to recognize an insignificant exertion, minimal size, capable and strong ac to dc converter to meet the data execution document of UPS. The execution of single stage and three stage ac to dc converter alongside different control systems are considered and thought about. This paper introduces a novel ac/dc converter in view of a semi dynamic power factor remedy (PFC) plot. In the proposed circuit, the power factor is upgraded by using bolster dc to dc converter. It wipes out the utilization of dynamic switch and control circuit for PFC, which brings about lower cost and higher productivity. A Matlab/Simulink based model is created and reenactment comes about are introduced. At last a DC engine stack is connected and reproduction comes about are displayed.

Keywords: AC/DC convereter; Power Factor Correction; Single stage.

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DVR Applications on Distribution Systems for Power Factor Improvement

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Keywords: Dynamic Voltage Restorer; Power Quality; Unit vector; Voltage harmonics; Voltage swell; \ .age sag.

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PAPER ID: 5034

AN OPTIMIZATION POWER OF ADDER TREES FOR MULTIPLE CONSTANT MULTIPLICATIONS

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Abstract

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Keywords: Railway track inspection, robot control, train control, gate control, Ultrasonic sensor, Infra-red sensor, GSM,GPS,RF transmitter and receiver

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Paper ID: 6010

A Literature Insception of MANET: History, Advantages,Challenges and its Applications

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Abstract

The migration to wireless network from wired networkhas been a global trend in the past few decades. The mobility and scalability brought by wireless network made it possible in many applications. Mobile Ad hoc NETwork (MANET) is one of the mostimportant and unique applications. On the contrary to traditional network architecture, MANET does not require a fixed network infrastructure; every single node works as both a transmitter and a receiver. Nodes communicate directly with each other when they are both within the same communication range. Otherwise, they rely on their neighbors to relay messages. While comapared to other networks .We present in this paper, the history of MANET, challenges (issues) involve in MANET ,Advantages and Disadvantages and some of its applications and their criteria.

Keywords: Mobile Ad hoc NETwork (MANET), Advantages,challenges,history

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Paper ID: 6013

Computer Network: Designing and Replication Of Load Balancing

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Abstract

The servers overload and the Quality of Service (QoS) results reduced and operation becomes more erious as the use of Web services is developed. In order to avoid this, service donors use huge administrator ıetworks of servers to attend the requests of the increasing visitors count in popular sites. OPNET (Optimum Network Performance) is used to develop a new model. The model was then evaluated to compute the peration of the wireless local area network. The model was used for two types of applications (ftp and ıttp) and found that among a set of other specifications response time and wireless media access delay vere highly affected by the number of users per application with and without load balancing. OPNET eplication showed the impact of load balancing on wireless and wire-line network for two different ypes of applications.

Keywords: WLAN, Load balancing, Media Access Delay, Http response time, ftp response time.

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Paper ID: 6018

End To End Encryption Based Privacy For Mental Health Care Data Transaction In The Cloud

Gangu Dharma Raju*, Adapa Chandrakala**, A V S Pawan Kumar***,M Prudhvi Ravi Raja Reddy****
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Abstract

From the prevailing procedure of uploading information and transmission by the reasons of its reasonable cost, fast, global scale, efficiency, behavior, and reliabilitythe Cloud computing is anextraordinary progress. The have no. of uses of cloud in the sector of education, social networking and medicine. But in medical the uses of cloud is ideal, particularly because the healthcare organizations data is produced very large. In cloud, as in increasingly health organizations adopting towards electronic health records which can be accessed around the world for different health problems regarding references, educational research in healthcare and etc. However, in the cloud data transferrable the privacy and security of data remain the demanding the problems. It is unavoidable which is stored and transferred using cloud to accept tight security to protect the integrity of the carefulmedical information, from the tampering attacks, hacking and intrusion.

Keywords: Cloud, Data Security, Integrity, Confidentiality, End to End Encryption, Multi-Factor Authentication, One-Time-Pad.

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Paper ID: 6020

The Practical Iterative MapReduce in Enormous Information Condition Completed on Hadoo

Mr. P Krishnanjaneyulu, Asst Professor1, Mr. B.S.Venkata Reddy, Associate Professor 2.
Mr. P Krishnanjaneyulu, Asst Professor1, Dept. of CSE, GIET Engineering College, Rajamahendravaram, AP. Email : krishna54888@gmail.com
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Abstract

Information is growing well ordered with the Improvement of Data Innovation. We could isolate mor imperative Data from the huge scale information. Directly a day's for all intents and purposes each onlin customers chase down things, organizations, purposes of intrigue et cetera to figure PageRank using th MapReduce way to deal with parallelization. This gives us a strategy for enlisting PageRank that can a basic level be therefore parallelized, in this way possibly scaled up to immense association graphs to far reaching aggregations of website pages. Delineate a single machine utilization which adequ handles a million or so pages. We use a gathering to scale out significantly assist – be interesting how far we can get. About PageRank and MapReduce at last overview the fundamental facts. We start with PageRank. Hadoop Guide Lessen is an item framework for viably making applications process impossible measures of information in parallel on broad clusters of thing gear in a str inadequacy tolerant way.

Keywords: Analysis, Big Data, Hadoop, Map Reduce, Report, Security.

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Paper ID: 6021

Video Steganography Using Lsb

P RAJANI DEVI, Baba Institute of technological sciences, Visakhapatnam. Email: rajaniperabathula@gmail.com.

bstract

he revolution in digital information has created new challenges for sending a message in a safe and ecure way. Whatever method we choose, the most important question is its degree of security. teganography is the art and science of invisible communication. Steganography plays an important role 1 field of Information Security. As for steganography cover types, almost any digital file format can be tilized for this purpose. Video is a very promising type of cover-media since it can carry a large amount f secret data. In addition, video steganography is becoming very important due to the frequent use and opularity of videos over the internet. Substitution-based techniques for implementing video teganography replace redundant data of the cover with the required secret message. Substitution-based echniques have numerous methods including the famous Least Significant Bit (LSB) technique, Bit 'lane Complexity Segmentation (BPCS), Tri-way Pixel Value Differencing (TPVD) etc., LSB is one of h oldest and most famous substitution-based techniques. In spite of its simplicity, it is capable of hiding arge secret messages. In this paper we will discuss video steganography with Least Significant bit ubstitution technique.

Keywords: Video Steganography; Secret message; Embedding; Extracting; Least significant bit method; Stego image.

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Paper ID: 6023

Cloud Computing-Data security using cryptographic technique to control Data Access Privileges in Cloud.

Maheswararao Kenguva, Ratna Kumar J
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Abstract

In the cloud computing environment, the data security plays a major role, because the data is placed in various locations across the world. Data security and isolation protection are the two main factors of user's concerns about the cloud computing technology. Data stored in the cloud to provide data confidentiality, data integrity, data access control, and data sharing. This paper is developing a system to control the data access by the unauthorized users. Using Google App Engine to provide various types of access controls and also protect the data for the users through cryptographic technique with minimal performance degradation the data should be securely stored in the public cloud service provider. This paper describes briefly about CCAF (Cloud Computing Adoption Framework) model for securing cloud data. This system is designed based on the necessities and the execution demonstrated by the CCAF multi-layered security. CCAF multi-layered security can be categorized into three layers to protect the data in real-time: 1) firewall and access control; 2) identity management and intrusion detection and prevention and 3) Encryption. CCAF can merge with the Auditing facility to record each and every action done by the user and system details of the registered users. CCAF also achieves to identify de-duplication is performed on a single file based on their hash values. The hash numbers are relatively easy to generate. Hence it requires less processing power.

Keywords: Isolation; Confidentiality; Integrity; data sharing; de-duplication; CryptographicTechnique CCAF.

Paper ID: 6025

Analysis of Crime data using data mining

K.S.N.MURTHY*, A.V.S.PAVAN KUMAR**, GANGU DHARMA RAJU***

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Abstract

Frequent pattern (itemset) mining plays a key role in rule mining. The Apriori and FP-developme calculations are the most renowned calculations which can be utilized for Frequent Pattern mining. h paper shows the study of different Frequent Pattern Mining and Rule Mining calculation which can applied to crime crime pattern mining. The analysis of survey would provide the data about what h been done already in a same area, what is the present trend and what are the other related areas. Freque Pattern mining having three important approaches that is candidate generation approach, without candida generation approach and vertical layout approach. It also explains different frequent pattern calculatio and how it can be connected to various zones especially in wrongdoing design identification. This pap

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Keywords: Frequent Pattern Mining, Apriori, FPgrowth, Association Rule Mining, Crime Pattern mining.

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Paper ID: 6026

Moving Objects patterns in Data Mining

A.V.S. Pavan Kumar*, G. Dharma Raju**, K.S.N.Murthy***

A.V.S.PAVAN KUMAR, Asst. Prof, Dept. of CSE, Baba Institute of Technology & Sciences, Visakhapatnam, AP, India.

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Abstract

Propelled innovation in GPS and sensors empowers us to track moving items, for example, people, creatures, vehicles. Action information of moving items, in some degree can mirror some inner and outside highlights of moving articles, how to utilize the monstrous high-exactness portable information recognize potential and significant example is the current problem areas and is additionally a difficult ... Patterns mining have various applications in human versatility understanding, urban arranging and natural investigations and in different fields. In this paper we shows a general point of idea to understand the techniques and calculations in depth which are occurs when we test the issues of pattern mining and comparing the existing results on the same issues. It helpful to the research worker to understand the issues.

Keywords: Patterns mining, Frequent Pattern Mining, Periodic Patterns Mining, following patterns mining

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Paper ID: 6027

Intranet Based Exam And Feedback Interface

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Paper ID: 6028

Placements Information For An Educational Instituion Through Web Portal

Medisetti Snehadivya*, Dakineni Durga Prasad**,Sanaka Kavitha***

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Placements information is a web based totally application advanced within the home windows platfoi for the training and placements branch of the college so as to offer the details of its college students ir database for the businesses to their process of recruitment furnished with a proper login. The placemei data carries all of the information about the students. The gadget stores all the personal information of t scholars, like their personal details, their combination marks, their talent set and their technical capabiliti which are required in the cv to be dispatched to a company. The device is an internet application that m be accessed at some stage in the company and outdoor as nicely with right login supplied. This devi can be used as an utility for the tpo of the college to control the pupil statistics close to placements. T project includes all the info of the students that can be considered by way of all of the users, but can changed handiest with the aid of the pupil with authorized service. by using maintaining student's statisti the machine enables to have choices to be made easy for a employer in its take a look at for the recruitm manner. the scholars can update their very own statistics handiest. students can search for the mater required for the selection method together with aptitude, reasoning and so forth. Events happening in college and the achievements of the student's i.e., selected scholar's information may be viewed by us all the customers. So, our challenge provides a facility of maintaining the information of the schol and receives the requested list of candidates for the agencies who would love to recruit the people ba at the given question.

Keywords: Curriculumvitae, Admin, Recruiter, Mailing, Material

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Paper ID: 6030

identifying objects in hazardous locations by cam robot using raspberry pi and iot technology

Gedela Sireesha Devi, Mr. K. Satya Narayana Murthy, Allu Gowri Sankar, Sireesha Devi Gedela, Email: srishamca@gmail.com

Abstract

Internet of Things have received significant attention during these days mainly because of their numerous applications. They can be found in robotics, smart buildings, fabrication equipments as well as medical, aautomation, industrial, commercial, military applications. Most of the modern Internet of Things based on Raspberry pi and Auridino boards and use it for constructing mobile robots. By interfacing a Wireless and Cam module into the proposed IoT system, it can provide a control system that uses Wireless as a standard technology for connecting remote devices. The control circuit of the proposed system is designed in a way such that a user can interface any kind of desirable peripheral. Furthermore, the cam robot is provided with a broadcast ip will open a web-page which has video screen for surveillance and buttons to control robot and camera.

Keywords: Internet of Things, Raspberry pi, Raspbian OS,putty,wireless networks,python programming,Usb cam, Robot kit,motion tool.

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IDENTIFYING OBJECTS IN HAZARDOUS LOCATIONS BY CAM ROBOT USING RASPBERRY PI AND IOT TECHNOLOGY

Gedela Sireesha Devi*
Mr. K. Satya Narayana Murthy**
Allu Gowri Sankar***

Abstract

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Sireesha Devi Gedela,
She is a student of Baba Institute of Technology and Sciences, Visakhapatnam. Presently she is pursuing her M.Tech from this college and she received her MCA from Gayatri College of PG Courses, Srikakulam. Her area of intrest includes Internet of Things, App Development and Python Programmig.

1.Introduction

Recent years have witnessed a remarkable growth in the features and applications of Internet of Things. Specifically, advances in IoT with computational capacity ready to support sophisticated intelligent mechanisms and the ability to host an overabundance of sensors, cams hence providing a new degree of freedom to sophisticated applications. Internet of Things usually stand for the integration of a smart system and its software into a larger system, which often monitors and controls equipment without the need for manual intervention. In other words, they are computingsystems that have computer hardware, software and internet embedded into them to perform computationally predefined tasks repeatedly on using significant amounts of application-specific fixed function logic. They can be found wide variety of devices such as Smart watches, Smart phones, Smart Electricals, Smart microwave ovens, Smart Home Security systems, Smart Scanners, Smart Home Appliances as well as other applications (e.g. Agriculture, medical, military etc.,).

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**He has a dedicated teaching experience of Eight years. He completed B.Tech from GVPCOE, Vizag and M.Tech from Andhra University,Now he was deeply involved in doing Research, Areas of Research interests includes Data mining, Image Processing, Computer Networks & Cloud Computing
*** Completed M.tech in Embedded systems, Continue research in IOT and realtime projects on embedded systems.

Paper ID: 6032

Applications of Computer Science Based on Graph theory

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Abstract

The field of mathematics plays an essential role in different domains. In various areas of graph theory employed in several applications of technology. Graphs are considered as an outstanding modeling t which is utilized to model several classes of relations amongst any physical scenario. Several issues the real world can be represented by graphs. This paper explores varied ideas concerned in graph the and their applications in computer science to demonstrate the utilization of graph theory. These applicati are given particularly to project the thought of graph theory and to demonstrate its objective and importar in computer science engineering.

Keywords: Graphs, Graph Theory, Computer Science and Applications

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Paper ID:7001

Manager's Task To Support Integrated Autonomy At The Workplace

Dr. Sreenivasa Rao Behara, Professor, MBA Department , BABA Institute of Technology and Sciences (BITS),PM Palem, Visakhapatnam- 500 048
Mr P.B. Narendra Kiran, HOD ,MBA Department , BABA Institute of Technology and Sciences (BITS),PM Palem, Visakhapatnam- 500 048

Abstract

A new managerial task arises in today's working life: to provide conditions for and influence interaction between actors and thus to enable the emergence of organizing structure in tune with a changing environment. We call this the enabling managerial task. The goal of this paper is to study whether training first line managers in the enabling managerial task could lead to changes in the work for the subordinates. This paper presents results from questionnaires answered by the subordinates of the managers before and after the training.The training was organized as a learning network and consisted of eight workshops carried out over a period of one year Each workshop consisted of three parts, during three and a half hours. The first hour was devoted to joint reflection on a task that had been undertaken since the last workshop; some results were presented from the employee pre-assessments, followed by relevant theory and illuminating practices, finally the managers created new tasks for themselves to undertake during the following month.The improvements are significant in scales measuring the relationship between the manager and the employees, as well as in those measuring interaction between employees. It is concluded that the result was a success for all managers that had the possibility of using the training in their management work.

Keywords: integration, autonomy, manager, enabling, communication, dialogue, improvisation.

Paper ID:7001

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Paper ID:7002

The Role Of Women Entrepreneurs As A Change Agent In The Society: A Case Study

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MR P.B. Narendra Kiran,HOD ,MBA Department , BABA Institute of Technology and Sciences (BITS),PM Palem, Visakhapatanam- 500048

Abstract

A society that does not have an optimistic, positive empowering image of the future is endangered. When society is skewed & power, it begins an entropic spiral to isolation – social entrepreneurs can use that skills to rebalance society. Society entrepreneurs are change agents that improve society by developing effective and equitable new models often less hierarchical yet more cooperative and complex than existing ones. Researchers and practitioners have defined society entrepreneurship in various ways but a common demonstrator is a venture that adds value to a community mission through innovative, risk-taking, business-like practices. Society entrepreneurship combines innovation with the importance of community needs and their work overlaps with social justice and environmental presentation movements. Women in Ancient India were in charge of the home affairs and were accorded very high status almost equal to that of man in religious duties. Gradually due to the varying social-political situation, women were relegated to the background and were subjected to exploitation.

Keywords: empowerment, entrepreneur, innovation.

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Keywords: empowerment, entrepreneur, innovation.

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Paper ID:7009

Training And Development Practices In Select Corporate Retail Stores In Visakhapatnam

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Abstract

A profound training program acts as a vehicle to enhance employee skills and enable them to perform better in their job. Training and development is very crucial to the employees, the organization and their effectiveness. Training is important in retailing because majority of the employees have direct contact with customers. They are responsible for helping customers, satisfy their needs and resolve their problems. Training is important not only from the point of view of employees, but also for the organization. This study has examined the perception of respondents on the present training and development practices of select corporate retail organizations. With this background, this study is conducted to know the perception of respondents in terms of their demographic profiles of gender, age, education and social status.

Keywords: Training, Development, Promotion

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Paper ID:7013

Operational And Financial Performance of Urban Cooperative Banks In India

Dr. P. Sanjeevi, Professor, Department of Management Studies, BITS Visakhapatnam
Mr. P. Manoj Babu, Asst. Professor, Department of Management Studies, BITS Visakhapatnam

Abstract

This study attempts to measure the operational and financial performance of Urban Cooperative Banks in India. The analytical frame of the operational tools in terms of the Credit Deposit Ratio and Investment Deposit Ratio, and two different financial segments, purposes; and also, the trends over the period under study have been presented. The study is based on secondary data collected from annual reports of urban cooperative banks in India for the study period i.e. 2005-06 to 2015-16. Ratios are used here with financial ratio analysis (FRA) method which helps to draw an overview about the financial performance of the urban cooperative banks in terms of profitability, liquidity and credit performance. To test the hypothesis the study has been worked on Statistical tools, correlation, t-test by using SPSS. These analyses are helped to see the current performance, condition of these urban banks to compare past performance. The study findings can be helpful for managing of urban cooperative banks. The study also identified specific areas for urban cooperative banks to work on which can ensure sustainable growth and development for these scheduled banks and non-scheduled banks.

Keywords- Operational Performance, Profitability, Financial ratio analysis, Credit Deposits Ratio and Investment Ratio.

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Paper ID:7019

Productivity Performance of The Indian Foreign Banks: An Experimental Analysis

Dr. P. Sanjeevi* Ms. Y. Sunitha** Mrs. G. Ch.V.Uma Rani***

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Abstract

Indian banking system has transformed in recent years due to globalization in the world market, whi has resulted in fierce competition. In this paper, an attempt has been made to analyze the productivi performance of Foreign Banks in India. Analytical frame of the operational performance per employ in terms of various tools Deposit, Advances, Total business, Total expenses, Establishment, Net pro and Investment, and to different financial segments, purposes; and also the trends over the period und study have been presented. The study is based on secondary data collected from annual reports of fi selected foreign banks in India for the study period i.e. 2004-05 to 2015-16. Ratios, t-test, correlati analysis have been used to analyze the present data. The study suggested that foreign banks in Ind should emphasize on generating more profits per employee by efficient utilization of its capital, tot investments, advances, deposits, debt and improving its operational efficiency and productivi performance.

Keywords: Total Business, Deposits per employee Investment per employee, productivity performanc

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Paper ID:7019

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Paper ID:7023

A Comparative Study on Work-Life Balance between Married and Unmarried Female Teachin Fraternity in an Engineering College

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Abstract

Work life balance plays a dominant role in an individual life and it was a challenging task especially f women to manage both the work life and personal life in this modern era. The present research stu identifies the causes of imbalance of work and personal life among the married and unmarried fema teaching fraternity in selected engineering college. This paper analyzes the causes of work and li imbalance with reference to female faculty. A total of 52 faculty responses of a selected engineeri college are included in the study. Statistical Analysis reveals that cause of imbalance of work life balan is spending more hours to travel to reach the work place, carrying work to home, not having enough tin

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Paper ID:7026

Consumer Attitude Towards The Products Of Hyper Markets In Visakhapatnam

Dr. B. Rama Jyothi, Assistant Professor; Mr. M. Vasudeva Rao, Assistant Professor; Mr. Naresh Kumar Kuppli, Assistant Professor

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Abstract

Market is a place where buyers and sellers meet, goods and services are offered for sale and transfer of ownership occurs. Sometimes a market is a precisely physical place and sometimes it refers to groups of people. At present day's hyper markets plays an important role to develop the modern markets and improve the customer services and promote the economic excellaration. As market consists of consumers, their attitude, beliefs, loyalty and perception is important in modern marketing. Without customers, a firm has no revenues, no profits, and therefore no market value. Consumers all have different tastes, like and dislikes and adopt different behaviour patterns while making purchase decisions. The present study deals with the attitude, consumer satisfaction and preferences with regard to the services, prices, quality, products etc. of hyper markets.

Keywords: Modern Markets; Customer Attitude; Customer satisfaction; Hyper markets;

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Keywords:Modern Markets; Customer Attitude; Customer satisfaction; Hyper markets;

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Paper ID:7028

First Year Student Expectations Of An Undergraduate Management Program: A Survey

Dr. G N P V Babu, Associate Professor, Baba Institute of Technology and Sciences, Visakhapatnam, Andhra Pradesh/
Mr. Rajesh Dorbala, Assistant Professor, Mittal School of Business, Lovely Professional University, Phagwara, Punjab

Abstract

Although much has been written on the first-year experience of students at higher education institutions, less attention has been directed to the expectations of students when they enter an institution for the first time. This paper delivers acumens into the expectations of first year students at largest private university in India and highlight areas in which students' expectations may not essentially line up with the realities of common university practices. By providing chances for students to coherent their expectations, staff are able to use the responses for a fruitful dialogue and work towards a more positive alignment between perceived expectations and levels of student satisfaction with their experience. It highlights that the expectations of the students are beyond traditional teaching methods and it is inclined towards career planning and orientation.

Keywords: Student Perception, Career Orientation, Traditional Teaching, Skill Set

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